THE LOCUS WHERE A MODULE IS COHEN-MACAULAY IS OPEN AND ITS CLOSURE IS THE SUPPORT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-dualizing/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The statement that a module over a local ring with vanishing local cohomology of the dual in all but one degree is Cohen-Macaulay (and conversely). Invoked inside the counted proof (source lines 3893-3898). Identifies the locus with the set of points where the dual has a unique nonzero cohomology module.
- Finiteness of the cohomology modules of the dual and the vanishing criterion for finitely generated modules. Used inside the counted proof (source lines 3899-3902). Produces a localizing element killing the unwanted cohomology.

Source reference. Original lemma lemma-local-CM: sources/stacks-dualizing/pinned-original-source.tex, source lines 3307–3307. Navigation pointer only: the counted text refers to the Stacks item lemma-local-CM, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Lemma 0.1. *Let A be a Noetherian ring with dualizing complex $\omega_A^\bullet$. Let M be a finite A-module. Then*

$$U = \{\mathfrak{p} \in \mathrm{Spec}(A) \mid M_\mathfrak{p} \text{ is Cohen-Macaulay}\}$$

is an open subset of $\mathrm{Spec}(A)$ *whose intersection with* $\mathrm{Supp}(M)$ *is dense.*

Proof. If $\mathfrak{p}$ is a generic point of $\mathrm{Supp}(M)$, then $\mathrm{depth}(M_\mathfrak{p}) = \dim(M_\mathfrak{p}) = 0$ and hence $\mathfrak{p} \in U$. This proves denseness. If $\mathfrak{p} \in U$, then we see that

$$R\mathrm{Hom}_A(M, \omega_A^\bullet)_\mathfrak{p} = R\mathrm{Hom}_{A_\mathfrak{p}}(M_\mathfrak{p}, (\omega_A^\bullet)_\mathfrak{p})$$

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has a unique nonzero cohomology module, say in degree i_0, by Lemma lemma-local-CM. Since $R\mathrm{Hom}_A(M, \omega_A^\bullet)$ has only a finite number of nonzero cohomology modules H^i and since each of these is a finite A-module, we can find an $f \in A$, $f \notin \mathfrak{p}$ such that $(H^i)_f = 0$ for $i \neq i_0$. Then $R\mathrm{Hom}_A(M, \omega_A^\bullet)_f$ has a unique nonzero cohomology module and reversing the arguments just given we find that $D(f) \subset U$. $\square$